

VICHY-CHARMEIL AP, France

WMO: 073740

Lat: 46.1667N

Lon: 3.3986E

Elev: 817

StdP: 14.27

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.4	22.7	13.7	11.4	20.9	18.4	14.3	25.7	18.9	44.6	16.7	43.7	2.3	190	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	92.3	69.2	88.0	68.0	84.2	66.7	71.1	86.4	69.5	83.9	68.0	81.1	5.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	100.7	76.1	64.9	95.0	73.9	63.6	90.6	72.3	35.3	86.6	33.9	83.8	32.7	81.2	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	13.9	11.9	12.9	98.4	6.0	4.3	8.6	101.5	5.0	103.9	1.7	106.3	-2.7	109.4
			12.3	74.6	6.0	1.8	8.0	75.8	4.5	76.9	1.2	77.9	-3.1	79.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	38.6	39.6	45.4	50.7	57.8	65.0	68.3	67.6	61.2	54.8	45.3	39.8	(6)
(7)		DBStd	12.68	7.71	7.89	6.69	6.15	6.34	6.61	6.04	6.15	5.92	7.35	7.24	7.64	(7)
(8)		HDD50	1438	360	298	170	65	8	0	0	0	2	38	175	322	(8)
(9)		HDD65	4790	818	712	607	428	237	80	33	40	142	322	591	780	(9)
(10)		CDD50	2506	6	5	28	88	250	451	567	545	338	187	34	7	(10)
(11)		CDD65	386	0	0	0	0	14	82	135	120	29	6	0	0	(11)
(12)		CDH74	4786	0	0	2	42	248	1036	1598	1406	392	61	1	0	(12)
(13)		CDH80	1853	0	0	0	3	58	405	662	619	100	6	0	0	(13)
(14)	Wind	WSAvg	4.5	4.6	5.3	5.4	5.2	4.5	4.4	4.3	3.8	3.6	3.8	4.3	4.7	(14)
(15)	Precipitation	PrecAvg	30.9	1.9	1.6	1.7	2.6	3.3	3.0	3.1	3.2	2.8	2.8	2.9	2.0	(15)
(16)		PrecMax	37.9	4.0	3.4	4.3	5.9	5.9	7.3	7.2	7.6	5.9	5.8	5.5	5.0	(16)
(17)		PrecMin	21.4	0.5	0.5	0.4	0.4	0.9	1.1	0.7	1.0	0.9	0.7	1.3	0.6	(17)
(18)		PrecStd	4.4	0.9	0.7	0.8	1.4	1.3	1.5	1.4	1.6	1.3	1.3	1.0	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.9	64.4	71.6	79.2	85.9	96.0	96.7	99.7	88.6	80.1	68.3	61.6	(19)
(20)		0.4%	MCWB	52.1	51.1	55.3	60.6	66.2	69.9	68.8	69.1	65.6	63.6	57.6	53.1	(20)
(21)		2%	DB	56.5	59.1	66.8	73.7	80.7	89.3	91.4	92.3	82.7	75.2	63.8	57.8	(21)
(22)		2%	MCWB	50.5	49.5	52.8	58.0	64.4	67.9	69.4	69.1	65.2	62.6	55.5	51.0	(22)
(23)		5%	DB	53.6	55.4	62.7	69.2	75.8	84.6	87.3	87.1	78.7	71.1	60.3	55.0	(23)
(24)		5%	MCWB	48.1	47.5	51.5	55.9	61.8	67.3	68.4	68.0	63.8	60.8	53.8	49.5	(24)
(25)		10%	DB	50.4	52.2	58.8	64.6	71.8	80.1	83.1	82.1	74.5	67.2	56.8	51.8	(25)
(26)		10%	MCWB	45.9	46.0	49.8	53.7	60.3	65.8	66.7	66.6	62.7	59.0	51.8	47.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.6	52.3	56.7	62.3	68.5	73.5	73.1	72.6	69.3	65.9	59.0	54.0	(27)
(28)		0.4%	MCDWB	58.9	60.8	67.0	75.4	81.3	88.0	88.3	89.8	80.2	75.7	65.9	59.1	(28)
(29)		2%	WB	51.1	50.2	54.2	59.1	65.7	70.6	70.8	70.4	67.0	63.6	56.5	51.9	(29)
(30)		2%	MCDWB	55.7	56.9	63.8	70.9	77.1	84.7	86.4	86.4	77.9	72.3	61.9	56.7	(30)
(31)		5%	WB	48.5	48.4	52.3	56.6	63.2	68.5	69.2	69.0	65.2	61.5	54.5	49.9	(31)
(32)		5%	MCDWB	52.8	54.6	60.8	66.6	72.9	81.2	84.0	83.4	75.6	69.2	59.6	54.2	(32)
(33)		10%	WB	46.3	46.5	50.5	54.4	61.0	66.6	67.6	67.3	63.3	59.7	52.2	47.7	(33)
(34)		10%	MCDWB	50.0	51.6	57.8	63.1	69.6	77.7	80.7	79.8	72.7	66.2	56.3	51.5	(34)
(35)	Mean Daily Temperature Range	MDBR	13.8	16.7	20.9	22.3	21.7	22.8	23.3	23.6	22.8	19.4	15.3	13.8	(35)	
(36)		5% DB	MCDDBR	17.6	23.5	29.7	31.7	29.7	31.9	31.5	33.0	30.4	25.1	21.3	18.4	(36)
(37)			MCWBR	12.1	15.2	17.6	17.3	14.8	13.6	12.5	12.7	14.3	13.6	13.8	12.7	(37)
(38)			MCDDBR	16.1	21.3	25.2	28.6	25.5	27.4	28.1	28.1	25.7	22.6	19.3	16.2	(38)
(39)		5% WB	MCWBR	11.6	14.3	15.5	16.1	13.6	12.4	11.7	11.8	12.9	12.9	12.9	11.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.325	0.354	0.384	0.390	0.398	0.393	0.388	0.361	0.353	0.321	0.302	(40)	
(41)		taud	2.481	2.422	2.358	2.280	2.314	2.326	2.334	2.360	2.412	2.437	2.482	2.484	(41)	
(42)		Ebn at Noon	254	269	276	277	278	276	275	272	271	255	245	242	(42)	
(43)		Edh at Noon	21	27	33	39	40	39	38	36	31	27	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	378	641	1031	1395	1634	1903	1859	1629	1259	794	440	322	(44)	
(45)		RadStd	48	94	124	176	156	174	143	137	89	62	48	54	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (25 neighbors)	(m) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page